RUTUJA JABRAS

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SUMMARY

Final-year B.Tech Computer Science student experienced in building full-stack, data-driven and AI-enabled applications using Python, SQL, React, REST APIs and machine learning. Focused on risk analysis, decision support and intelligent business workflows, with a strong problem-solving mindset and interest in scalable enterprise technologies.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript
Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Access Control
Frontend: React.js
Databases: MySQL, MongoDB, SQLAlchemy (ORM)
AI / ML: XGBoost, SHAP, RAG Pipelines (ChromaDB, TF-IDF), NLP
Tools & Platforms: Git, Cloudinary, Vercel, Render, Pydantic
Core CS: Data Structures & Algorithms (Arrays, Strings, Trees, Recursion), Problem Solving

PROJECTS

Burnout & Attrition Intelligence System | *2026*

- Built a full-stack HR analytics platform combining XGBoost and SHAP for explainable attrition risk prediction with a RAG pipeline over HR policy documents grounded in the Job Demands–Resources (JD-R) model.
- Architected a FastAPI backend with JWT authentication, SQLAlchemy ORM, Pydantic schemas, and role-based access control for secure, enterprise-style access.
- Designed a React frontend featuring a custom JD-R strain gauge visualization to translate model output into interpretable psychological insights.

KisaanSaathi – Farmer Loan Distress Early Warning System | *2025*

- Developed a hybrid RAG architecture (ChromaDB + TF-IDF + Groq LLaMA3-8B) to assess farmer loan distress risk using real Indian government scheme data.
- Built a deterministic hybrid risk engine and a scheme-gap scoring system to flag at-risk farmers and match them to relevant welfare schemes.
- Implemented the backend in Flask with a React/Vite frontend for an accessible, low-latency user experience.

AuditX – Financial Risk Analysis System | *2026*

- Developed a machine learning system to analyze vendor risk in government fund allocation scenarios using predictive models built with XGBoost.
- Designed a web interface for monitoring risk scores and managing complaints, improving visibility into high-risk entities.

CiviCare AI – Civic Complaint Management System | *2026*

- Built a fully deployed civic complaint platform with OTP authentication, AI-based urgency classification via Groq, and image uploads via Cloudinary.
- Used MongoDB for data storage and deployed the system end-to-end across Vercel (frontend) and Render (backend).
- Resolved a live production security issue involving an exposed API key, reinforcing secure credential management practices.

EXPERIENCE

Web Development Intern – Sadhana IT Solutions Pvt. Ltd. | *June 2025 – July 2025*

- Completed hands-on training in full-stack development using React.js, Node.js, and MongoDB.
- Built and tested REST APIs and developed responsive frontend components.

EDUCATION

B.Tech in Computer Science and Engineering, MIT-ADT University | *2023–2027*
CGPA: 7.4

Class XII, Maharashtra State Board – 80% | *2023*

Class X, Maharashtra State Board – 98.40% | *2021*

ACHIEVEMENTS & CERTIFICATIONS

- Participated in AI Grand Challenge at MIT-ADT University.
- Certifications: Generative AI, Data Analytics, and Machine Learning.